

Welcome to my Data Analysis Internship journey at NeoSkillz. This document represents my learning, practical work, and hands-on experience in the field of data analytics. It includes important concepts, notes, assignments, and techniques I explored while working with Python, SQL, data visualization, and data analysis tools. This collection reflects my growth, dedication, and interest in transforming data into meaningful insights.

Data Analysis Internship

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DAY 1 – INTRODUCTION TO DATA ANALYTICS

Data Analytics is the process of collecting, cleaning, transforming, and analyzing data to generate useful business insights. It helps organizations understand patterns, trends, and relationships within data to support better decision-making.

The main objective of analytics is not only to discover insights but also to use them for improving business performance, increasing efficiency, and achieving organizational goals.

Through data analytics, raw data is converted into meaningful information that helps companies solve problems, improve strategies, increase profitability, and make effective data-driven decisions.

DATA ANALYTICS WORKFLOW

Step 1

- Raw Data Collection :- Data is collected from different sources. It can be in unstructured format and needs proper organisations

Step 2

- Data Structure :- Raw data is converted into structured format eg - CSV ,JSON ,XML

STEP 3

- Data Analysis :- data is analysed to find any pattern, relationships, business insights and trends

Step 4

- Insights Generation :- Analysis results are converted into simple business terms .These insights help in making effective decisions

Step 5

- Decision Making :- Data-Driven decisions help improves business performance and profitability.

Business Intelligence Approach in Data Analytics

A **Business Intelligence (BI) approach in Data Analytics** is a process of collecting, organizing, analyzing, and presenting data to help businesses make better decisions. It transforms raw data into meaningful insights that support planning, strategy, and performance improvement.

Key steps in a BI approach include:

1. **Data Collection** – Gathering data from various internal and external sources.
2. **Data Cleaning & Preparation** – Removing errors, handling missing values, and preparing data for analysis.
3. **Data Analysis** – Finding trends, relationships, and insights from data.
4. **Data Visualization** – Creating dashboards, charts, and reports for easy understanding.
5. **Decision Making** – Using insights to improve business strategies and outcomes.

Business Intelligence helps organizations improve efficiency, understand customers, track performance, reduce risks, and make data-driven decisions.

Case Study

Trading journal Analysis

- Situation:** Two traders , Vijendera and Nithin, achieved the same net profit (+9 points) . however, their risk profile were different.
- Analysis:** Statistical analysis shows that vijendra FOMO trade reduced his return by 2 points.
- Insights:** Removing taht single mistake could make his profile from +9 to +11 points.
- Learning:** Data analysis helps identify problems and improve future performance.

E-Commerce Smart Pricing

- Problem:** Reducing product price to compete with competitors can decrease profit margins.
- Data-Deiven Solutions:** Parterning with banking companies for card cashback offers. The bank margin most of the discount cost. The companies maintains its profit margin while attracting consumers
- Learning:** Data helps business make smarter decisions and stratagies insted of only copying competitors.